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| Inventor(s) | Kitahara; Nobuyasu et al. |


## PROCESSING METHOD


### Abstract

A processing method includes forming a recess circumferentially along a support plate including a face having a size equal to or larger than the diameter of a device region and supporting a device wafer, in an outer circumferential region of the face that is positioned outwardly of a central region thereof that is commensurate with the device region, affixing the face of the support plate and the face side of the device wafer to each other with use of at least resin supplied to the recess, thinning down the device wafer by grinding or polishing or by grinding and polishing a reverse side of the device wafer, removing a region of the resin circumferentially along the device wafer, thereby to reduce a bonding force between the device wafer and the support plate, and separating a portion of the device wafer that includes the device region from the support plate.



**Inventors:** **Kitahara; Nobuyasu (Tokyo, JP), Chen; Zhiwen (Tokyo, JP), Teranishi; Shunsuke (Tokyo, JP), Oda; Atsushi (Tokyo, JP)**

**Applicant:** **DISCO CORPORATION** (Tokyo, JP)



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## Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a processing method of processing a device wafer having on its face side a device region where devices have been formed and an excess outer circumferential region surrounding the device region.

Description of the Related Art

[0002] Electronic appliances such as cellular phones and personal computers (PCs) incorporate device chips. Device chips are manufactured as follows, for example. A wafer of monocrystalline silicon that has a plurality of devices such as integrated circuits (ICs) constructed on its face side is thinned down by having its reverse side ground by a grinding apparatus. Then, the thinned wafer is divided by a cutting apparatus into device chips including the respective devices.

[0003] In recent years, there have been demands in the art for wafers that have been thinned down to a thickness in a range from 50 μm to 100 μm. However, wafers having a thickness equal to or smaller than 100 μm are more difficult to handle than wafers having a thickness larger than 100 μm because they are more susceptible to damage during delivery after being thinned down.

[0004] In view of the above difficulty, there has been known in the art a process of grinding the reverse side of a wafer while the face side, i.e., a device side, of the wafer is affixed to a support plate of glass, also referred to as a support substrate or a carrier wafer, for example, by an organic adhesive such as an ultraviolet-curable resin (see, for example, JP 2015-222755A).

[0005] After the reverse side of the wafer has been ground, the support plate is peeled off from the wafer after ultraviolet rays have been applied to the adhesive to reduce its bonding strength. While being delivered after the thinning process, the wafer is prevented from being damaged, and the thinned-down wafer peeled off from the support plate can be processed in a subsequent process.

[0006] However, the method disclosed in JP 2015-222755A is problematic in that the adhesive tends to remain unremoved on the face side of the thinned-down wafer peeled off from the support plate. It has been proposed in the art to form an annular recess, i.e., a bottomed annular slot, in an outer circumferential region of the support plate that is aligned with an excess outer circumferential region of the wafer, supply the annular recess with an adhesive, and affix the wafer and the support plate to each other with the adhesive (see, for example, JP 2013-026614A).

[0007] If the adhesive used in the process disclosed in JP 2013-026614A includes an ultraviolet-curable resin, then it is necessary to apply ultraviolet rays to the adhesive in order to reduce the bonding strength of the adhesive at the time the wafer is to be peeled off from the support plate.

[0008] However, since wafers of monocrystalline silicon are almost impermeable to ultraviolet rays, material constraints are imposed on the support plate because it needs to be made of a material such as glass that is permeable to ultraviolet rays, instead of monocrystalline silicon.

[0009] If a thermosetting resin is used as the adhesive, then it would make it possible to use another process called a laser debonding process that reduces the bonding strength of the adhesive by irradiating the adhesive with a laser beam in an ultraviolet wavelength band. However, the laser beam in the ultraviolet wavelength band still imposes material constraints on the wafer carrier because the wafer carrier needs to be made of a material such as glass that is permeable to ultraviolet rays.

SUMMARY OF THE INVENTION

[0010] The present invention has been made in view of the foregoing problems. It is an object of the present invention to provide a processing method that reduces constraints on the materials of an adhesive and a support plate providing after a wafer and the support plate have been affixed to each other by using an adhesive, the wafer is separated from the support plate.

[0011] In accordance with an aspect of the present invention, there is provided a processing method of processing a device wafer having on a face side thereof a device region where devices have been formed and an excess outer circumferential region surrounding the device region. The method includes forming a recess circumferentially along a support plate including a face having a size equal to or larger than a diameter of the device region and supporting the device wafer, in an outer circumferential region of the face that is positioned outwardly of a central region thereof that is commensurate with the device region, or circumferentially along the device wafer in the excess outer circumferential region of the face side of the device wafer, supplying the recess with resin, affixing the face of the support plate and the face side of the device wafer to each other with use of at least the resin, thereby forming a stack, thinning down the device wafer by grinding or polishing or by grinding and polishing a reverse side of the device wafer that is positioned opposite the face side of the device wafer in a thicknesswise direction thereof, removing a region of the resin that contributes to the affixing of the device wafer and the support plate to each other circumferentially along the device wafer, thereby to reduce a bonding force with which the device wafer and the support plate are bonded to each other in a thicknesswise direction of the stack, or forming an annular slot in the device wafer that extends from the face side to the reverse side thereof outwardly of the device region and inwardly of an annular region aligned with the recess, thereby to sever a portion of the device wafer that includes the device region from the support plate, and separating the portion of the device wafer that includes the device region from the support plate.
[0012] Preferably, the processing method further includes, before the stack is formed, forming an additional recess circumferentially in the excess outer circumferential region or the outer circumferential region, which is free of the recess, of the face of the support plate and the face side of the device wafer, and before the stack is formed, supplying the additional recess with an additional resin.
[0013] Preferably, in supplying the recess with the resin, while a lower end of a cutting blade is positioned at a first depth from the reverse side of the device wafer or at a second depth from another face of the support plate that is positioned opposite the face of the support plate in a thicknesswise direction thereof, the region of the resin that contributes to the affixing of the device wafer and the support plate to each other is removed by the cutting blade circumferentially along the device wafer, thereby reducing the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.
[0014] In supplying the recess with the resin, a pulsed layer beam having a wavelength absorbable by the device wafer may be applied from the reverse side of the device wafer to another face of the support plate that is positioned opposite the face of the support plate in a thicknesswise direction thereof or from the other face of the support plate to the reverse side of the device wafer, thereby to remove the region of the resin that contributes to the affixing of the device wafer and the support plate to each other circumferentially along the device wafer, thereby reducing the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.
[0015] In thinning down the device wafer, a central portion of the reverse side of the device wafer that is commensurate with the device region may be thinned down to form a ring-shaped stiffener on an outer circumferential portion thereof that is commensurate with the excess outer circumferential region, and in reducing the bonding force with which the device wafer and the support plate are bonded to each other or in severing the portion of the device wafer from the support plate, the region of the resin that contributes to the affixing of the device wafer and the support plate to each other may be removed circumferentially along the device wafer to reduce the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.
[0016] Preferably, while the support plate is held on a holding table and the device wafer is held on a holding plate different from the holding table, the bonding force with which the device wafer and

the support plate are bonded to each other is reduced or the portion of the device wafer is severed from the support plate.
[0017] Preferably, the support plate includes a plurality of pores extending therethrough from the central region of the face thereof along a thicknesswise direction of the support plate to another face of the support plate that is positioned opposite the face thereof, and in forming the stack, the face of the support plate and the face side of the device wafer are affixed to each other.
[0018] In the processing method according to the aspect of the present invention, since the face of the support plate and the face side of the device wafer are affixed to each other using the resin supplied to the recess in the outer region of the support plate or in the excess outer circumferential region of the face side of the device wafer, there is no adhesive left in the device region of the device wafer peeled off from the support plate.
[0019] Moreover, the processing method includes removing the region of the resin that contributes to the affixing of the device wafer and the support plate to each other, thereby to reduce the bonding force with which the device wafer and the support plate are bonded to each other, or forming the annular slot in the device wafer that extends from the face side to the reverse side thereof outwardly of the device region and inwardly of the annular region aligned with the recess, thereby to sever the portion of the device wafer that includes the device region from the support plate.
[0020] Consequently, the support plate can be peeled off from the device wafer without applying a laser beam having an ultraviolet wavelength band through the support plate to an adhesive. The adhesive and the support plate are thus less liable to suffer material constraints than with the related art.
[0021] The above and other objects, features and advantages of the present invention and the manner of realizing them will become more apparent, and the invention itself will best be understood from a study of the following description and appended claims with reference to the attached drawings showing some preferred embodiments of the invention.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] FIG. **1** is a flowchart of a processing method according to a first embodiment of the present invention;
[0023] FIG. **2**A is a perspective view of a device wafer;
[0024] FIG. **2**B is a perspective view of a support plate;
[0025] FIG. **3** is an enlarged fragmentary cross-sectional view taken along line A-A of FIG. **2**B, illustrating an outer circumferential edge portion of the support plate;
[0026] FIG. **4** is a perspective view illustrating a recess forming step of the processing method according to the first embodiment;
[0027] FIG. **5** is a side elevational view, partly in cross section, illustrating the recess forming step;
[0028] FIG. **6**A is a cross-sectional view of the support plate that has undergone the recess forming step;
[0029] FIG. **6**B is a side elevational view, partly in cross section, illustrating a resin supplying step of the processing method according to the first embodiment;
[0030] FIG. **7**A is a side elevational view, partly in cross section, illustrating an affixing step of the processing method according to the first embodiment;
[0031] FIG. **7**B is a side elevational view, partly in cross section, of a stack according to the first embodiment;
[0032] FIG. **8**A is a perspective view illustrating a thinning step of thinning down the device wafer by way of in-feed grinding;

[0033] FIG. **8**B is a cross-sectional view of the device wafer that has been ground;
[0034] FIG. **9** is a perspective view illustrating a thinning step of thinning down the device wafer by way of polishing;
[0035] FIG. **10** is a side elevational view, partly in cross section, illustrating a resin removing step in a separation preparing step of the processing method according to the first embodiment;
[0036] FIG. **11** is a side elevational view, partly in cross section, illustrating a separating step;
[0037] FIG. **12** is a side elevational view, partly in cross section, illustrating a resin removing step according to a first modification of the first embodiment;
[0038] FIG. **13** is a side elevational view, partly in cross section, illustrating a recess in a support plate according to a second modification of the first embodiment;
[0039] FIG. **14**A is a perspective view illustrating a recess forming step of a processing method according to a second embodiment of the present invention;
[0040] FIG. **14**B is a side elevational view, partly in cross section, illustrating a resin supplying step of the processing method according to the second embodiment;
[0041] FIG. **15** is a flowchart of a processing method according to a third embodiment of the present invention;
[0042] FIG. **16**A is a side elevational view, partly in cross section, illustrating an affixing step of the processing method according to the third embodiment;
[0043] FIG. **16**B is a side elevational view, partly in cross section, of a stack according to the third embodiment;
[0044] FIG. **17** is a side elevational view, partly in cross section, illustrating an annular groove forming step in a separation preparing step of a processing method according to a fourth embodiment of the present invention;
[0045] FIG. **18**A is a side elevational view, partly in cross section, illustrating the manner in which a laser beam is applied in a resin removing step of a processing method according to a fifth embodiment of the present invention;
[0046] FIG. **18**B is a side elevational view, partly in cross section, illustrating the manner in which the laser beam is applied in the resin removing step of the processing method according to a modification of the fifth embodiment;
[0047] FIG. **19** is a side elevational view, partly in cross section, illustrating a thinning step of a processing method according to a sixth embodiment of the present invention;
[0048] FIG. **20** is a side elevational view, partly in cross section, illustrating a stack that has undergone the thinning step of the processing method according to the sixth embodiment;
[0049] FIG. **21** is a side elevational view, partly in cross section, illustrating a resin removing step in a separation preparing step of the processing method according to the sixth embodiment;
[0050] FIG. **22**A is a plan view of a support plate according to a modification; and
[0051] FIG. **22**B is an enlarged fragmentary cross-sectional view taken along line B-B of FIG. **22**A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

First Embodiment

[0052] A processing method according to a first embodiment of the present invention will be described in detail below with reference to the accompanying drawings. FIG. **1** is a flowchart of the processing method according to the first embodiment for processing a device wafer **11** (see FIG. **2**A). As illustrated in FIG. **1**, the processing method according to the first embodiment includes step S**10** of forming a recess (recess forming step), step S**20** of supplying the recess with resin (resin supplying step), step S**30** of forming a stack (i.e., a laminated assembly) **25** (see FIG. **7**B) (affixing step), step S**40** of thinning down the device wafer **11** (thinning step), step S**50** of reducing the bonding strength between the device wafer **11** and a support plate **21** (see FIG. **2**B) or separating part of the device wafer **11** from the support plate **21** (separation preparing step), and step S**60** of separating part of the device wafer **11** including a device region from the support plate

**21** (separating step). These steps are carried out in the order named.

[0053] The steps of the processing method according to the first embodiment will be described below. First, the device wafer **11** will be described below with reference to FIG. **2**A. FIG. **2**A illustrates the device wafer **11** in perspective. The device wafer **11** has a disk-shaped, i.e., plate-shaped, substrate made of monocrystalline silicon (Si), also referred to as a silicon wafer. The substrate of monocrystalline silicon has a thickness ranging from approximately 250 μm to approximately 800 μm depending on the diameter thereof.

[0054] The substrate of the device wafer **11** is not limited to any particular materials, types, sizes among others. For example, the device wafer **11** may have a monocrystalline substrate made of any of other semiconductor materials than silicon that include gallium nitride (GaN), silicon carbide (SiC), diamond, and gallium oxide (Ga.sub.2O.sub.3).

[0055] The device wafer **11** has a face side **11***a* with a grid of projected dicing lines or streets **13** established thereon along which the device wafer **11** will be divided. The projected dicing lines **13** demarcate a plurality of rectangular areas on the face side **11***a* where respective devices **15** such as ICs are constructed. The devices **15** are not limited to any particular types, numbers, shapes, structures, sizes, and layouts, for example.

[0056] According to the present embodiment, the face side **11***a* includes a central region referred to as a device region **17***a*. The device region **17***a* includes a circular region on the face side **11***a* that contains all the devices **15** and that extends radially outwardly but terminates short of an outer circumferential edge portion of the device wafer **11**.

[0057] According to the present embodiment, furthermore, the face side **11***a* also includes an annular region radially surrounding the device region **17***a* and extending to an outer circumferential edge of the device wafer **11**. The annular region is referred to an outer circumferential edge portion **17***b*. In FIG. **2**A, an annular boundary **17***c* between the device region **17***a* and the outer circumferential edge portion **17***b* is indicated by the broken line. The boundary **17***c* is depicted only for illustrative purposes in FIG. **2**A and actually not plotted on the face side **11***a*.

[0058] The device wafer **11** has a reverse side **11***b* opposite the face side **11***a* across its thickness along a thicknesswise direction **11***c*. The substrate of monocrystalline silicon is exposed on the reverse side **11***b*. Prior to thinning step S**40** of thinning down the device wafer **11** by grinding or polishing or by grinding and polishing the reverse side **11***b*, the device wafer **11** and the support plate **21** (see FIG. **2**B) are affixed to each other.

[0059] FIG. **2**B illustrates the support plate **21** in perspective. The support plate **21** has a disk-shaped substrate of monocrystalline silicon, i.e., a silicon wafer, such as a dummy wafer, a test wafer, or a mirror wafer. The support plate **21** is free of devices **15**. The material of the support plate **21** is not limited to a substrate of monocrystalline silicon and may be any of other materials than monocrystalline silicon, including ceramic, glass, resin, or metal, for example.

[0060] The material of the support plate **21** may be either permeable or impermeable to ultraviolet rays or radiations in other wavelength bands than ultraviolet rays. According to the present embodiment, material constraints imposed on the support plate **21** are thus reduced, and material constraints imposed on an adhesive used to bond the device wafer **11** and the support plate **21** to each other are reduced.

[0061] The support plate **21** is not limited to any particular thicknesses, but should preferably have a thickness in excess of 100 μm, e.g., a thickness of 1.0 mm, to make itself rigid to a certain degree. The support plate **21** has a circular face **21***a* and another circular face **21***b* that are opposite each other across its thickness along a thicknesswise direction **21***c*.

[0062] The face **21***a* has a diameter, i.e., a size, equal to or larger than the diameter of the device region **17***a* of the device wafer **11**. According to the present embodiment, the diameter of the face **21***a* is substantially the same as the diameter of the face side **11***a*. However, if the support plate **21** has a rectangular substrate, then each of the sides of the face **21***a* may be equal to or larger than the diameter of the device region **17***a* in order to enable the face **21***a* to keep the face side **11***a* in its

entirety thereon.
[0063] The face **21***a* has a circular central region **23***a* that is commensurate with the device region **17***a*. The central region **23***a* is equal in diameter to the device region **17***a*. According to the present embodiment, an annular region of the face **21***a* that is positioned radially outwardly of the central region **23***a* and extends radially outwardly to an outer circumferential edge **23***e* (see FIG. **3**) of the support plate **21** is referred to as an outer circumferential region **23***b*.
[0064] In FIG. **2**B, an annular boundary **23***c* between the central region **23***a* and the outer circumferential region **23***b* is indicated by the broken line in FIG. **2**B only for illustrative purposes. Actually, the boundary **23***c* is not plotted on the support plate **21**.
[0065] FIG. **3** illustrates, in enlarged fragmentary cross section, an outer circumferential edge portion of the support plate **21**, taken along line A-A of FIG. **2**B. In FIG. **3**, the outer circumferential edge **23***e* of the support plate **21**, a position **23***d* on the face **21***a* that is spaced radially inwardly from the outer circumferential edge **23***e* by a predetermined distance of 1 mm, for example, and the boundary **23***c* are also depicted.
[0066] An annular region extending radially outwardly from the boundary **23***c* to the predetermined position **23***d* is less planar than the central region **23***a* of the face **21***a*, and the shape of the annular region is called "edge roll-off." According to the present embodiment, the annular region is referred to as a "roll-off region **23***f*."
[0067] A distance **23***g* that extends along the thicknesswise direction **21***c* from the predetermined position **23***d* to an imaginary plane, indicated by the broken line in FIG. **3**, parallel to the face **21***a* of the support plate **21** is occasionally used as an index representing the planarity of the roll-off region **23***f* (roll-off amount: ROA).
[0068] According to the present embodiment, another annular region extending radially outwardly from the predetermined position **23***d* to the outer circumferential edge **23***e* is referred to as a "beveled region **23***h*." The roll-off region **23***f* and the beveled region **23***h* jointly make up the outer circumferential region **23***b*. The outer circumferential edge portion **17***b* of the face side **11***a* of the device wafer **11** is also of a cross-sectional shape similar to the cross-sectional shape of the roll-off region **23***f* and the beveled region **23***h*. Annular regions of the reverse side **11***b* that correspond to the outer circumferential edge portion **17***b* along the thicknesswise direction **11***c* are of a similar cross-sectional shape.
[0069] In a case where the support plate **21** has a substrate of monocrystalline silicon such as a dummy wafer, for example, the support plate **21** usually has the roll-off region **23***f* and the beveled region **23***h*. However, support plates **21** made of other materials and/or having other shapes may not necessarily have the roll-off region **23***f* and the beveled region **23***h*.
[0070] Recess forming step S**10** will be described below with reference to FIGS. **4** and **5**. FIG. **4** illustrates recess forming step S**10** in perspective, and FIG. **5** illustrates recess forming step S**10** in side elevation, partly in cross section. An X-axis extending horizontally that is indicated by an arrow X, a Y-axis extending horizontally that is indicated by an arrow Y, and a Z-axis extending vertically that is indicated by an arrow Z, are illustrated in FIGS. **4** and **5** and extend perpendicularly to each other.
[0071] In recess forming step S**10** according to the present embodiment, a cutting apparatus **2** is used to form an annular recess **21***d* extending circumferentially in the roll-off region **23***f* of the support plate **21**. The cutting apparatus **2** includes a disk-shaped chuck table **4** having a disk-shaped frame made of metal.
[0072] The frame has a cavity defined diametrically centrally therein that is smaller in diameter than the frame. A disk-shaped porous plate, not depicted, made of porous ceramic is fixed fitted in the cavity. The frame and the porous plate have respective upper surfaces that are exposed upwardly and lie substantially flush with each other, jointly providing a substantially flat upper holding surface **4***a* (see FIG. **5**).
[0073] The holding surface **4***a* lies substantially parallel to an XY plane that is defined along the X-

axis and the Y-axis. The porous plate is fluidly connected to a suction source, not depicted, such as a vacuum pump, for example. When the suction source is actuated, it generates and transmits a negative pressure to the porous plate, thereby creating a suction force on the holding surface **4***a*. The chuck table **4** has a lower surface whose central portion is fixed to the upper end of a rotational shaft **4***b* (see FIG. **5**) extending substantially parallel to the Z-axis.

[0074] The rotational shaft **4***b* has a lower end connected to a rotary actuator, not depicted, such as an electric motor. When the rotary actuator is energized, it rotates the rotational shaft **4***b* about its vertical central axis, rotating the chuck table **4** about its central axis aligned with the vertical central axis of the rotational shaft **4***b*. The chuck table **4** and the rotational shaft **4***b* are movable in unison horizontally along the X-axis by a moving mechanism, not depicted, that includes a ball screw.

[0075] The cutting apparatus **2** also includes a cutting unit **6** disposed above the holding surface **4***a* of the chuck table **4**. The cutting unit **6** has a cylindrical spindle **8** (see FIG. **5**) whose longitudinal central axis extends horizontally along the Y-axis. The spindle **8** has a proximal end portion, not depicted, connected to rotary actuator, not depicted, such as an electric motor. The spindle **8** has a distal end portion on which there is mounted a cutting blade **10** having an annular cutting edge **10***a*. When the rotary actuator is energized, it rotates the spindle **8** about its longitudinal central axis and hence the cutting blade **10** about its horizontal central axis aligned with the longitudinal central axis of the spindle **8**, rotating the cutting edge **10***a* about its central axis aligned with the horizontal central axis of the cutting blade **10**. The cutting unit **6** is movable along the Y-axis and the Z-axis by a moving mechanism, not depicted, that includes a ball screw.

[0076] The cutting apparatus **2** is controlled in operation by a controller, not depicted. The controller includes a computer including a processor, typically, a central processing unit (CPU), a main storage device such as a dynamic random access memory (DRAM), and an auxiliary storage device such as a solid-state drive. The auxiliary storage device stores software including certain programs. The processor is operated according to the software to enable the controller to perform its functions.

[0077] In recess forming step S**10**, as illustrated in FIGS. **4** and **5**, the face **21***a* of the support plate **21** is exposed upwardly and the other face **21***b* thereof is put in contact with the holding surface **4***a* of the chuck table **4**. Then, the support plate **21** is held under suction on the chuck table **4** by the suction force created on the holding surface **4***a*.

[0078] Thereafter, the spindle **8** is rotated about its longitudinal central axis at a predetermined rotational speed, and the cutting unit **6** is lowered from above the support plate **21**, causing the cutting edge **10***a* to cut into the roll-off region **23***f* on the face **21***a*, i.e., to make a chopper cut. At this time, the cutting edge **10***a* has a lower end **10***b* whose position is appropriately adjusted depending on the depth of the recess **21***d* to be formed in the roll-off region **23***f*. The recess **21***d* has a width in the radial directions of the face **21***a* that is equal to the thickness of the cutting edge **10***a*.

[0079] The rotational shaft **4***b* is turned about the vertical central axis thereof through a predetermined angle per unit time. At least while the chuck table **4** makes one revolution, the cutting unit **6** is fixed in position vertically along the Z-axis, cutting the support plate **21** to form the recess **21***d* to a predetermined depth in the roll-off region **23***f*.

[0080] During the cutting process, cutting water such as pure water is supplied at a predetermined flow rate to an area where the cutting edge **10***a* and the support plate **21** are kept in contact with each other. An example of cutting conditions is given below. The width of the recess **21***d* may be made larger than the thickness of the cutting edge **10***a* by changing the position of the cutting unit **6** along the Y-axis with respect to the holding surface **4***a* between the first revolution of the chuck table **4** and the second or subsequent revolution thereof. [0081] Rotational speed of cutting blade: 20000 rpm [0082] Angle through which chuck table is turned per unit time: 5°/s [0083] Flow rate of cutting water: 5 L/min [0084] Depth of recess: 0.04 mm to 0.40 mm from vertical position of central region **23***a* [0085] Width of recess: 1.0 mm to 3.0 mm

[0086] Instead of making the chopper cut, the chuck table **4** may be moved along the X-axis while the lower end **10***b* of the rotating cutting edge **10***a* is fixed in position along the Z-axis, causing the cutting edge **10***a* to cut to a desired position in the roll-off region **23***f*, after which the chuck table **4** may start rotating about its vertical central axis.

[0087] Resin supplying step S**20** will be described below. FIG. **6**A illustrates, in cross section, the support plate **21** that has undergone recess forming step S**10**. After recess forming step S**10**, an affixing apparatus **12** (see FIG. **6**B) is used to supply a liquid resin **19** to the recess **21***d* (resin supplying step S**20**). The affixing apparatus **12** has a disk-shaped chuck table **14** (see FIG. **6**B).

[0088] The chuck table **14** is essentially identical in structure and shape to the chuck table **4** of the cutting apparatus **2**, and will be omitted from detailed description. The chuck table **14** has an upper holding surface **14***a* and a rotational shaft **14***b*.

[0089] As illustrated in FIG. **6**B, the affixing apparatus **12** includes a dispenser unit **16** disposed above the chuck table **14**. The dispenser unit **16** includes a needle **18** for discharging the liquid resin **19**. The liquid resin **19** is discharged from the needle **18** at a flow rate controlled by a controller, not depicted, of the dispenser unit **16**. The controller includes an electric circuit, not depicted.

[0090] The controller outputs an electric signal for controlling the opening and closing of a valve, not depicted, on a pipe interconnecting a tank, not depicted, that is filled with compressed air and a syringe, not depicted, filled with the liquid resin **19**. The controller controls the timing of opening and closing of the valve thereby to control the rate at which the liquid resin **19** is discharged from the needle **18**. Alternatively, while the liquid resin **19** that has filled a tank is delivered under pressure to the valve, the controller may control the opening and closing of the valve thereby to control the rate at which the liquid resin **19** is discharged from the needle **18**.

[0091] According to the present embodiment, the liquid resin **19** includes a thermosetting epoxy resin. However, the liquid resin **19** may include another resin such as an acrylic resin, for example. The liquid resin **19** is not limited to a thermosetting epoxy resin but may be resin **19** that is curable by way of solvent evaporation or exposure to ultraviolet rays.

[0092] FIG. **6**B illustrates resin supplying step S**20** in side elevation, partly in cross section. In resin supplying step S**20**, while the needle **18** has its lower opening facing the recess **21***d*, the liquid resin **19** is supplied from the opening of the needle **18** to the recess **21***d* at the same time that the chuck table **14** is turned about its vertical central axis through a predetermined angle per unit time.

[0093] The liquid resin **19** is supplied to fill the recess **21***d* preferably to the extent that the liquid resin **19** will finally be slightly convex, i.e., slightly protrusive, from the face **21***a* of the support plate **21**. The total amount of the liquid resin **19** thus supplied is appropriately adjusted to affix the face side **11***a* of the device wafer **11** and the face **21***a* of the support plate **21** to each other without the liquid resin **19** reaching the device region **17***a* when the face side **11***a* is held against the face **21***a*.

[0094] After resin supplying step S**20**, the chuck table **14** is moved to a predetermined position by a moving mechanism. After the position of the support plate **21** has been finely adjusted using a camera in that position, a suction holding unit, not depicted, that is holding the device wafer **11** under suction is lowered as illustrated in FIG. **7**A.

[0095] As described above, the liquid resin **19** includes a thermosetting epoxy resin according to the present embodiment. After the face side **11***a* has contacted the liquid resin **19** in the recess **21***d*, a heat source, not depicted, such as a resistance heater that may be embedded in the chuck table **14** is energized to cure the liquid resin **19**. When the liquid resin **19** is cured, it bonds the device wafer **11** and the support plate **21** to each other.

[0096] FIG. **7**A illustrates affixing step in side elevation, partly in cross section. FIG. **7**B illustrates the stack **25** produced in affixing step S**30** in side elevation, partly in cross section. As illustrated in FIG. **7**B, the face **21***a* of the support plate **21** and the face side **11***a* of the device wafer **11** are affixed to each other by at least the cured resin **19**, making up the stack **25**.

[0097] In the stack **25**, the device wafer **11** is supported on the support plate **21**. Even when the device wafer **11** is thinned down subsequently in thinning step S**40**, the stack **25** remains rigid enough because of the support plate **21** included therein. According to the present embodiment, after affixing step S**30**, a grinding and polishing apparatus **22** (see FIGS. **8**A, **8**B, and **9**) is used to grind and polish the reverse side **11***b* of the device wafer **11**, thereby thinning down the device wafer **11**.

[0098] The grinding and polishing apparatus **22** will be described below with reference to FIGS. **8**A, **8**B, and **9**. An X-axis extending horizontally that is indicated by the arrow X, a Y-axis extending horizontally that is indicated by the arrow Y, and a Z-axis extending vertically that is indicated by the arrow Z, are illustrated in FIGS. **8**A, **8**B, and **9** and extend perpendicularly to each other. The grinding and polishing apparatus **22** includes a disk-shaped chuck table **24** having a disk-shaped frame made of dense ceramic.

[0099] The frame has a cavity defined diametrically centrally therein that is smaller in diameter than the frame. A disk-shaped porous plate, not depicted, made of porous ceramic is fixed fitted in the cavity. The frame and the porous plate have respective upper surfaces that are exposed upwardly and lie substantially flush with each other, jointly providing a substantially flat upper holding surface **24***a* (see FIG. **8**B). The holding surface **24***a* is of an upwardly protruding conical shape including a central portion that protrudes slightly upwardly beyond an outer circumferential portion thereof by a very small distance ranging from approximately 10 μm to approximately 30 μm. In FIG. **8**B, therefore, the holding surface **24***a* is illustrated essentially flatwise.

[0100] The porous plate is fluidly connected to a suction source, not depicted, such as a vacuum pump, for example. When the suction source is actuated, it generates and transmits a negative pressure to the porous plate through the frame, thereby creating a suction force on the holding surface **24***a*. The chuck table **24** has a lower surface whose central portion is fixed to the upper end of a rotational shaft **24***b* extending substantially parallel to the Z-axis.

[0101] The rotational shaft **24***b* has a lower end connected to a rotary actuator, not depicted, such as an electric motor. When the rotary actuator is energized, it rotates the rotational shaft **24***b* about its vertical central axis, rotating the chuck table **24** about its central axis aligned with the vertical central axis of the rotational shaft **24***b*. The rotational shaft **24***b* is slightly inclined such that a portion of the holding surface **24***a* that is commensurate with a processed region of the device wafer **11** lies substantially parallel to an XY plane that is defined along the X-axis and the Y-axis. The grinding and polishing apparatus **22** includes a grinding unit **26** (see FIG. **8**A) disposed above the holding surface **24***a*.

[0102] The grinding unit **26** has a cylindrical spindle **28** whose longitudinal central axis extends vertically along the Z-axis. A rotary actuator, not depicted, such as an electric motor is coupled to a longitudinally central portion of the spindle **28**. The spindle **28** has a lower end to which a disk-shaped wheel mount **30** is fixed.

[0103] A disk-shaped grinding wheel **32** is mounted on the lower surface of the wheel mount **30**. The grinding wheel **32** has an annular base **32***a* of metal that has a lower surface on which there is mounted an annular array of grindstones **32***b* spaced at substantially equal angular intervals circumferentially along the base **32***a*. The grinding and polishing apparatus **22** further includes a polishing unit **34** (see FIG. **9**) disposed near the grinding unit **26**.

[0104] The polishing unit **34** has a cylindrical spindle **36** whose longitudinal central axis extends vertically along the Z-axis. A rotary actuator, not depicted, such as an electric motor is coupled to a longitudinally central portion of the spindle **36**. The spindle **36** has a lower end to which a disk-shaped wheel mount **38** is fixed.

[0105] A disk-shaped polishing tool **40** is mounted on the lower surface of the wheel mount **38**. The polishing tool **40** has a disk-shaped base **40***a* of metal that has a lower surface to which a polishing pad **40***b* is fixed. The polishing pad **40***b* includes a polishing pad of bonded abrasive grains. For example, the polishing pad **40***b* has a base member made of foamed hard polyurethane, for example, and abrasive grains made of silica, for example, bonded to and in the base member.

[0106] The polishing tool **40** has a through hole, not depicted, defined diametrically centrally therein and extending through the base **40***a* and the polishing pad **40***b* along the Z-axis. While the grinding and polishing apparatus **22** is polishing the reverse side **11***b* of the device wafer **11**, a polishing liquid suitable for the polishing of the workpiece to be polished, i.e., the device wafer **11**, is supplied via the through hole to the reverse side **11***b*.
[0107] Thinning step S**40** will be described below with reference to FIGS. **8**A, **8**B, and **9**. FIG. **8**A illustrates, in perspective, thinning step S**40** of thinning down the device wafer **11** by way of in-feed grinding. In thinning step S**40**, the face **21***b* of the support plate **21** is held under suction on the holding surface **24***a* such that the face **21***a* of the support plate **21** is exposed upwardly.
[0108] Then, the chuck table **24** is positioned directly below the grinding unit **26**, after which the rotational shaft **24***b* and the spindle **28** are rotated in respective directions and the grinding unit **26** is lowered, i.e., grinding-fed, along the Z-axis at a predetermined speed. While the grinding unit **26** is grinding the device wafer **11**, grinding water such as pure water is supplied at a predetermined flow rate to an area where the grindstones **32***b* and the reverse side **11***b* are kept in contact with each other.
[0109] The grinding process in thinning step S**40** includes, for example, a coarse grinding process using a coarse grinding wheel that includes coarse grindstones as the grindstones **32***b* and a fine grinding process using a fine grinding wheel that includes fine grindstones as the grindstones **32***b*.
[0110] The coarse grindstones have a granularity of #320, for example, whereas the fine grindstones have a granularity of #2000, for example. The granularity represents the size of the abrasive grains. For details of the granularity, reference should be made to JIS R 6001-2:2017 (the granularity of grinding materials for grindstones—part 2: fine powder) of the Japanese Industrial Standards (JIS).
[0111] An example of grinding conditions is given below. The length of time during which the coarse grinding process is carried out and the length of time during which the fine grinding process is carried out are determined depending on the thickness of the device wafer **11** to be ground away.
Coarse Grinding Process:
[0112] Rotational speed of spindle: 3200 rpm [0113] Rotational speed of chuck table: 300 rpm
[0114] Grinding feed speed: 3.0 μm/s [0115] Flow rate of grinding water: 4.0 L/min
Fine Grinding Process:
[0116] Rotational speed of spindle: 3200 rpm [0117] Rotational speed of chuck table: 300 rpm
[0118] Grinding feed speed: 0.3 μm/s [0119] Flow rate of grinding water: 4.0 L/min
[0120] After the entire reverse side **11***b* has been coarsely ground, the fine grinding process is carried out. After the device wafer **11** has been thinned down to a desired thickness by the coarse grinding process and the fine grinding process, the grinding unit **26** is lifted along the Z-axis. FIG. **8**B illustrates, in cross section, the device wafer **11** that has been ground.
[0121] According to the present embodiment, the reverse side **11***b* that has been ground is polished. After the reverse side **11***b* has been ground, the chuck table **24** stops being rotated and is moved to a position directly below the polishing unit **34** (see FIG. **9**). FIG. **9** illustrates thinning step S**40** of thinning down the device wafer **11** by way of polishing.
[0122] During the polishing process, an alkaline aqueous solution is supplied as a polishing liquid at a predetermined flow rate to an area where the polishing pad **40***b* and the reverse side **11***b* are kept in contact with each other. At the same time, the polishing pad **40***b* is pressed against the reverse side **11***b* of the device wafer **11** on the chuck table **24** under a predetermined pressing force, thereby polishing the reverse side **11***b* in its entirety.
[0123] If no abrasive grains are included in the polishing pad **40***b*, then the polishing liquid contains abrasive grains. Depending on the material of the workpiece, i.e., the device wafer **11**, to be polished, the aqueous solution has its components selected and adjusted appropriately. For example, if the workpiece includes a substrate of monocrystalline silicon carbide, then an acidic aqueous solution is used as the polishing liquid. An example of polishing conditions according to

the present embodiment is given below. [0124] Rotational speed of spindle: 750 rpm [0125] Rotational speed of chuck table: 745 rpm [0126] Pressing force: 40 kPa [0127] Flow rate of polishing liquid: 200 mL/min [0128] Polishing period: 620 s

[0129] In thinning process S**40** according to the present embodiment, the device wafer **11** is both ground and polished. However, depending on the material and thickness of the device wafer **11**, the polishing process may be omitted and the device wafer **11** may be only ground or the grinding process may be omitted and the device wafer **11** may be only polished.

[0130] After thinning step S**40**, separation preparing step S**50** is carried out using the cutting apparatus **2** described above. Separation preparing step S**50** will be described below. According to the present embodiment, separation preparing step S**50** includes a step of removing the region of the cured resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other, thereby reducing the bonding force with which the device wafer **11** and the support plate **21** are bonded to each other along a thicknesswise direction **25***a* of the stack **25** (resin removing step).

[0131] Removing the region of the cured resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other does not necessarily mean removing the entire resin **19** completely. The region of the cured resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other can be removed by removing at least the portion of the resin **19** that is held in contact with the face side **11***a*.

[0132] By removing at least the portion of the resin **19** that is held in contact with the face side **11***a*, the main force involved in bonding the device wafer **11** and the support plate **21** to each other is cancelled. The thickness along the Z-axis of the portion of the resin **19** to be removed may be appropriately determined in view of the accuracy with which the cutting apparatus **2** is able to cut the workpiece and the period of time required for the cutting apparatus **2** to cut the workpiece.

[0133] In the resin removing step, the chuck table (holding table) **4** holds the face **21***b* of the support plate **21** under suction thereon, and a holding plate **42** different from the chuck table **4** holds the reverse side **11***b* of the device wafer **11** under suction thereon.

[0134] The holding plate **42** includes a frame of metal that is smaller in diameter than the chuck table **4**. The frame has a disk-shaped cavity, not depicted, defined in a lower surface thereof, and a disk-shaped porous plate, not depicted, is fixedly fitted in the cavity. The frame and the porous plate have respective lower surfaces lying substantially flush with each other, jointly providing a substantially flat holding surface **42***a*.

[0135] The holding surface **42***a* is substantially equal in diameter to the device region **17***a* or smaller in diameter than the device region **17***a*, and lies substantially parallel to the XY plane. The porous plate is fluidly connected to a suction source, not depicted, such as a vacuum pump, for example. When the suction source is actuated, it generates and transmits a negative pressure to the porous plate through the frame, thereby creating a suction force on the holding surface **42***a*.

[0136] A rotational shaft **42***b* extending substantially parallel to the Z-axis has a lower end fixed centrally to an upper surface of the holding plate **42**. The rotational shaft **42***b* has an upper end connected to a rotary actuator, not depicted, such as an electric motor. When the rotary actuator is energized, it rotates the rotational shaft **42***b* about its vertical central axis and hence the holding plate **42** about its vertical central axis aligned with the vertical central axis of the rotational shaft **42***b*. The holding plate **42** and the rotational shaft **42***b* are movable in unison with each other along at least the Z-axis by a moving mechanism, not depicted, including a ball screw.

[0137] FIG. **10** illustrates the resin removing step in separation preparing step S**50** in side elevation, partly in cross section. In the resin removing step according to the present embodiment, first, the face **21***b* of the support plate **21** is held under suction on the chuck table **4**, and the portion of the reverse side **11***b* of the device wafer **11** that is positioned radially inwardly of the cured resin **19** is held under suction on the holding plate **42**.

[0138] Then, the spindle **8** is rotated about its longitudinal central axis at a predetermined rotational speed, and the cutting unit **6** is lowered from above the support plate **21**, positioning the lower end **10***b* of the cutting edge **10***a* at a first depth C**1** extending from the reverse side **11***b* of the device wafer **11** along the Z-axis, i.e., to making a chopper cut.

[0139] Instead of making the chopper cut, the chuck table **4** may be moved along the X-axis while the lower end **10***b* of the rotating cutting edge **10***a* is fixed at the first depth C**1** along the Z-axis, causing the cutting edge **10***a* to cut to a desired position in the roll-off region **23***f*, after which the chuck table **4** may start rotating about its vertical central axis.

[0140] The position of the lower end **10***b* of the rotating cutting edge **10***a*, i.e., the first depth C**1**, is closer to the face **21***a* of the support plate **21** than at least the face side **11***a* of the device wafer **11**. According to the present embodiment, the lower end **10***b* of the cutting edge **10***a* is positioned within the support plate **21**.

[0141] With the cutting edge **10***a* thus cutting into the outer circumferential edge portion **17***b* of the device wafer **11** and the cured resin **19**, the chuck table **4** and the holding plate **42** start being rotated and are turned through the same angle per unit time.

[0142] At least while the chuck table **4** makes one revolution, the cutting unit **6** is fixed in position vertically along the Z-axis, removing the region of the resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other circumferentially along the device wafer **11**. In this manner, the region of the resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other is removed circumferentially along the device wafer **11** by the cutting blade **10**. The bonding force with which the device wafer **11** and the support plate **21** are bonded to each other along the thicknesswise direction **25***a* is now reduced.

[0143] At this time, since the device wafer **11** is held under suction on the holding plate **42**, the disk-shaped region as the device region **17***a* is prevented from being separated and scattered from the support plate **21**, and the device region **17***a* is prevented from being cut in error when the device wafer **11** is thus cut.

[0144] During the cutting process, cutting water such as pure water is supplied at a predetermined flow rate to an area where the cutting edge **10***a* and the stack **25** are kept in contact with each other. An example of cutting conditions is given below. [0145] Rotational speed of cutting blade: 20000 rpm [0146] Angle through which chuck table is turned per unit time: 5°/s [0147] Flow rate of cutting water: 5.0 L/min

[0148] In separation preparing step S**50** according to the present embodiment, the region of the cured resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other is removed without applying a laser beam in an ultraviolet wavelength band to the adhesive, i.e., the resin **19**, through the support plate **21**.

[0149] Consequently, in subsequent separating step S**60**, the support plate **21** can be peeled off from the device wafer **11**. As there are not constraints that would result from the transmission of ultraviolet rays through the support plate **21**, the adhesive and the support plate **21** are less liable to suffer material constraints than with the related art.

[0150] According to the present embodiment, the resin removing step in separation preparing step S**50** uses the cutting apparatus **2** that is used in recess forming step S**10**. However, a cutting apparatus including the chuck table **4**, the holding plate **42**, and the cutting unit **6** but different from the cutting apparatus **2** may be used in the resin removing step. In other words, either the cutting apparatus **2** or another cutting apparatus different from the cutting apparatus **2** may be used in the resin removing step. The other cutting apparatus may use another cutting blade having the same cutting edge thickness as the cutting blade **10** or another cutting blade having a larger cutting edge thickness than the cutting blade **10**.

[0151] The holding plate **42** may not necessarily be limited to the structure including the frame and the porous plate described above. Rather, the holding plate **42** may have a base member made of metal, resin, or ceramic and one or a plurality of suction pads disposed on a lower surface of the base member for supplying a negative pressure. The stack **25** may be turned upside down, so that the reverse side **11***b* of the device wafer **11** may be held under suction on the chuck table **4** and the face **21***b* of the support plate **21** may be held under suction on the holding plate **42**.

[0152] Separation preparing step S**50** (resin removing step) is followed by separating step S**60**. FIG. **11** illustrates separating step S**60** in side elevation, partly in cross section. In separating step S**60**, the holding plate **42** and the rotational shaft **42***b* are lifted in unison along the Z-axis, separating the remaining portion of the device wafer **11** including the device region **17***a* from the support plate **21**.

[0153] In separation preparing step S**50**, inasmuch as the main force involved in bonding the device wafer **11** and the support plate **21** to each other with the cured resin **19** has already been cancelled, when the remaining portion of the device wafer **11** is lifted, the device wafer **11** including the device region **17***a* can easily be separated, i.e., peeled off, from the support plate **21**.

First Modification

[0154] A first modification of the first embodiment will be described below with reference to FIG. **12**. According to the first modification, as illustrated in FIG. **12**, a chuck table **44** that is equal to or smaller in diameter than the device region **17***a* is used in a resin removing step. The chuck table **44** has an upper holding surface **44***a* with a rotational shaft **44***b* fixed centrally to a lower surface of the chuck table **44**.

[0155] FIG. **12** illustrates a resin removing step (separation preparing step S**50**) according to the first modification of the first embodiment. In the resin removing step according to the first modification, the central portion of the reverse side **11***b* of the device wafer **11** is held under suction on the chuck table **44**, and the central portion of the face **21***b* of the support plate **21** is held under suction on the holding plate **42**.

[0156] Then, the spindle **8** is rotated about its longitudinal central axis at a predetermined rotational speed, and the cutting unit **6** is lowered from above the support plate **21**, positioning the lower end **10***b* of the cutting edge **10***a* at a second depth C**2** extending from the face **21***b* of the support plate **21** along the Z-axis, i.e., to make a chopper cut. According to the first modification, the second depth C**2** is positioned beneath the holding surface **44***a* along the Z-axis.

[0157] With the cutting edge **10***a* thus cutting into the outer circumferential edge portion **17***b* of the device wafer **11** and the cured resin **19**, the chuck table **44** and the holding plate **42** are turned through the same angle per unit time, removing the region of the resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other circumferentially along the device wafer **11** with the cutting blade **10**.

[0158] By thus simultaneously cutting the outer circumferential region **23***b* of the support plate **21** and the outer circumferential edge portion **17***b* of the device wafer **11**, the outer circumferential edge portion **17***b* of the device wafer **11** and the outer circumferential region **23***b* of the support plate **21** are severed in the thicknesswise direction **25***a*. As a result, the bonding force with which the device wafer **11** and the support plate **21** are bonded to each other along the thicknesswise direction **25***a* of the stack **25** is reduced.

[0159] Instead of making the chopper cut, the chuck table **44** may be moved along the X-axis while the lower end **10***b* of the rotating cutting edge **10***a* is fixed in the second depth C**2** along the Z-axis, causing the cutting edge **10***a* to cut to a desired position in the roll-off region **23***f*, after which the chuck table **44** may start rotating about its vertical central axis. Moreover, the outer circumferential edge portion **17***b* and the outer circumferential region **23***b* may be cut away by a cutting blade, not depicted, having a cutting edge thickness that is substantially the same as the respective width of the outer circumferential edge portion **17***b* and the outer circumferential region **23***b*.

Second Modification

[0160] A second modification of the first embodiment will be described below with reference to FIG. **13**. According to the second modification, as illustrated in FIG. **13**, the recess **21***d* is expanded radially outwardly of the support plate **21** to form an annular step whose outer circumferential portion is open radially outwardly in the outer circumferential region **23***b* of the support plate **21** on the face **21***a* thereof.

[0161] FIG. **13** illustrates the recess **21***d* in the support plate **21** according to the second modification of the first embodiment. For example, the expanded recess **21***d* may be formed by cutting the support plate **21** with an edge-trimming cutting blade, not depicted, having a relatively large cutting edge thickness ranging from 2 mm to 4 mm, for example. Alternatively, the recess **21***d* that provides the annular step may be formed by adjusting the position along the Y-axis of the cutting unit **6** with respect to the holding surface **4***a* upon the first revolution of the chuck table **4** and the second or subsequent revolution thereof.

Second Embodiment

[0162] A second embodiment of the present invention will be described below with reference to FIGS. **14**A and **14**B. According to the second embodiment, a recess **17***d* extending circumferentially along the device wafer **11** is formed in the outer circumferential edge portion **17***b* of the face side **11***a* of the device wafer **11**, instead of the support plate **21**.

[0163] FIG. **14**A illustrates recess forming step S**10** according to the second embodiment in side elevation, partly in cross section. FIG. **14**B illustrates resin supplying step S**20** according to the second embodiment in side elevation, partly in cross section. In subsequent affixing step S**30**, the support plate **21** is lowered toward the device wafer **11**. However, as with the first embodiment, the device wafer **11** may be lowered toward the support plate **21**.

[0164] The recess **17***d* has a depth ranging from 0.04 mm to 0.09 mm from the face side **11***a* of the outer circumferential edge portion **17***b*, for example, and a width ranging from 1.0 mm to 3.0 mm, for example.

Third Embodiment

[0165] A third embodiment of the present invention will be described below with reference to FIGS. **15**, **16**A, and **16**B. According to the third embodiment, the recess **17***d* is formed in the face side **11***a* of the device wafer **11**, and the recess **21***d* is formed in the face **21***a* of the support plate **21**.

[0166] FIG. **15** is a flowchart of a processing method according to the third embodiment. As illustrated in FIG. **15**, before resin supplying step S**20** and affixing step S**30**, recess forming step S**10** is followed by step S**12** of forming an additional recess (additional recess forming step).

[0167] In recess forming step S**10**, the annular recess **21***d* is formed in the face **21***a*. In additional recess forming step S**12**, the recess **17***d* is formed in the face side **11***a*. However, the recess **17***d* may be formed in the face side **11***a* in recess forming step S**10**, and the recess **21***d* may be formed in the face **21***a* in additional recess forming step S**12**.

[0168] In other words, in the face **21***a* of the support plate **21** and the face side **11***a* of the device wafer **11**, the additional recess may be formed circumferentially in the outer circumferential edge portion **17***b* or the outer circumferential region **23***b* where no recess has been formed in recess forming step S**10**.

[0169] According to the third embodiment, the width of the recess **21***d* formed in the face **21***a* of the support plate **21** is larger than the width of the recess **17***d* formed in the face side **11***a* of the device wafer **11** (see FIGS. **16**A and **16**B).

[0170] For example, a cutting edge **10***a* having a relatively large thickness is used to form the recess **21***d* in the face **21***a*, and a cutting edge **10***a* having a relatively small thickness is used to form the recess **17***d* in the face side **11***a*. Alternatively, the recess **21***d* may be formed by moving the cutting edge **10***a* having the relatively small thickness in a plurality of repetitive cycles circumferentially along the support plate **21**.

[0171] In a case where the support plate **21** and the device wafer **11** are positioned such that the face **21***a* and the face side **11***a* have their centers aligned with each other, the recess **21***d* and the recess **17***d* are formed in superposed positions circumferentially fully along the stack **25** in the thicknesswise direction **25***a* of the stack **25** (see FIG. **16**B).

[0172] However, the present invention is not limited to the above details. In a case where the support plate **21** and the device wafer **11** are positioned such that the face **21***a* and the face side **11***a* have their centers aligned with each other, unless the recess **21***d* overlaps the device region **17***a*, the recess **21***d* and the recess **17***d* may partially overlap each other or may not fully overlap each other.

[0173] After recess forming step S**10** and additional recess forming step S**12** but before affixing step S**30**, resin supplying step S**20** is carried out to supply a recess, i.e., one of the recess **17***d* and the recess **21***d*, with the liquid resin **19**, and step S**22** of supplying an additional recess, i.e., the other of the recess **17***d* and the recess **21***d*, with the liquid resin **19** (additional resin supplying step) is carried out.

[0174] The resin used in additional resin supplying step S**22** may not contain completely identical components to those of the resin **19**. FIG. **16**A illustrates affixing step S**30** according to the third embodiment in side elevation, partly in cross section, and FIG. **16**B illustrates the stack **25** formed in affixing step S**30** according to the third embodiment.

[0175] After affixing step S**30**, the same steps as those according to the first embodiment or the first modification thereof are carried out. Therefore, those steps are omitted from description. The second modification of the first embodiment may be applied to the third embodiment.

Fourth Embodiment

[0176] A fourth embodiment of the present invention will be described below with reference to FIG. **17**. According to the fourth embodiment, in separation preparing step S**50**, the resin removing step of cutting the cured resin **19** is replaced with a step of forming an annular groove **17***e* (see FIG. **17**) extending from the face side **11***a* to the reverse side **11***b* in the outer circumferential edge portion **17***b* of the device wafer **11**, thereby separating the portion of the device wafer **11** including the device region **17***a* from the support plate **21** (annular groove forming step).

[0177] FIG. **17** illustrates the annular groove forming step in separation preparing step S**50** according to the fourth embodiment in side elevation, partly in cross section. In the annular groove forming step illustrated in FIG. **17**, the face **21***b* of the support plate **21** is held under suction on the chuck table **4**, and the central portion of the reverse side **11***b* of the device wafer **11** is held under suction on the holding plate **42**.

[0178] Then, the spindle **8** is rotated about its longitudinal central axis at a predetermined rotational speed, and the cutting unit **6** is lowered from above the device wafer **11**, positioning the lower end **10***b* of the cutting edge **10***a* at a position deeper than the face side **11***a* of the device wafer **11** along the Z-axis, i.e., to make a chopper cut.

[0179] At this time, the lower end **10***b* of the cutting edge **10***a* may be positioned between the face side **11***a* and the face **21***a* to minimize wear on the cutting edge **10***a* as it does not cut the support plate **21**. However, the lower end **10***b* of the cutting edge **10***a* may be positioned on the face **21***a* or may cut into a position in the support plate **21** that is deeper than the face **21***a*.

[0180] The lower end **10***b* of the cutting edge **10***a* that is positioned on the face **21***a* or in the position deeper than the face **21***a* is more reliably effective to avoid the situation in which the device wafer **11** and the support plate **21** would not be separated from each other because the annular groove **17***e* would not extend through the device wafer **11**, leaving an uncut region in the device wafer **11**.

[0181] Instead of making the chopper cut, the chuck table **4** may be moved along the X-axis while the lower end **10***b* of the rotating cutting edge **10***a* is fixed in position along the Z-axis, causing the cutting edge **10***a* to cut to a desired position in the roll-off region **23***f*, after which the chuck table **4** may start rotating about its vertical central axis.

[0182] In the annular groove forming step, with the cutting edge **10***a* thus cutting into the outer circumferential edge portion **17***b* of the device wafer **11**, the chuck table **4** and the holding plate **42** are turned through the same angle per unit time. In this manner, the annular groove **17***e* is formed in the device wafer **11** radially outwardly of the device region **17***a* of the device wafer **11** and radially inwardly of an annular region thereof that is aligned with the recess **21***d* of the support plate **21**, as viewed in plan with respect to the stack **25**.

[0183] The annular groove **17***e* makes it possible to separate a disk-shaped region of the device wafer **11**, i.e., a portion of the device wafer **11**, that includes the device region **17***a* from the support plate **21**. According to the fourth embodiment, less severe material constraints are imposed on the adhesive and the support plate **21** than with the related art.

Fifth Embodiment

[0184] A fifth embodiment of the present invention will be described below with reference to FIG. **18**A. According to the fifth embodiment, in separation preparing step S**50**, a laser processing apparatus **52** is used instead of the cutting apparatus **2** to perform the resin removing step.

[0185] The laser processing apparatus **52** will be described below with reference to FIG. **18**A. As illustrated in FIG. **18**A, the laser processing apparatus **52** has a chuck table **54** having an upper holding surface **54***a*. A rotational shaft **54***b* extending substantially parallel to the Z-axis has an upper end fixed centrally to a lower surface of the chuck table **54**.

[0186] The chuck table **54** is structurally identical to the chuck table **4**, and its redundant description will be omitted below. However, the chuck table **54** is movable along the X-axis and the Y-axis by a moving mechanism, not depicted, having ball screws.

[0187] The laser processing apparatus **52** also includes a laser beam applying unit **56** disposed above the chuck table **54**. The laser beam applying unit **56** is movable along the Z-axis by a moving mechanism, not depicted, having a ball screw. The laser beam applying unit **56** includes a laser oscillator, not depicted, for emitting a pulsed laser beam.

[0188] The laser oscillator has a laser medium of Nd:YAG or Nd:YVO.sub.4, for example. When the pulsed laser beam emitted from the laser oscillator passes through a nonlinear optical crystal in the pulsed laser beam applying unit **56**, the laser beam is converted into a pulsed laser beam L having a wavelength of 355 nm, for example, absorbable by the device wafer **11**, and the pulsed laser beam L is emitted from the laser beam applying unit **56**.

[0189] The laser beam L travels through an optical system, not depicted, including a mirror to an irradiating head **58**. The irradiating head **58** has a condensing lens, not depicted, that focuses the laser beam L onto a point positioned along the Z-axis.

[0190] FIG. **18**A illustrates, in side elevation, partly in cross section, the manner in which the laser beam L is applied to the stack **25** in the resin removing step according to the fifth embodiment. In the resin removing step, the laser beam L from the irradiating head **58** is applied to the device wafer **11** through which the laser beam L travels from the reverse side **11***b* toward the face **21***b* of the support plate **21**. The laser beam L that has traveled through the device wafer **11** is focused into a focused spot P that is positioned in the vicinity of the boundary between the face side **11***a* and the face **21***a*.

[0191] Then, the chuck table **54** and the holding plate **42** are turned through the same angle per unit time, removing the region of the resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other, by way of ablation circumferentially along the device wafer **11**.

[0192] The bonding force with which the device wafer **11** and the support plate **21** are bonded to each other along the thicknesswise direction **25***a* of the stack **25** is now reduced. As illustrated in FIG. **18**B, the stack **25** may be turned upside down in its vertical orientation. An example of laser processing conditions is given below. [0193] Wavelength of laser beam: 55 nm to 532 nm [0194] Average output power: 0.5 W [0195] Repetitive frequency: 40 kHz [0196] Speed at which focused spot is fed: 1000 mm/s to 3000 mm/s

Modification of Fifth Embodiment

[0197] A modification of the fifth embodiment will be described below with reference to FIG. **18**B. FIG. **18**B illustrates, in side elevation, partly in cross section, the manner in which the laser beam L is applied to the stack **25** in the resin removing step according to the modification of the fifth embodiment.

[0198] According to the modification of the fifth embodiment, the laser beam L is applied from the face **21***b* of the support plate **21** toward the reverse side **11***b* of the device wafer **11**, and its focused

spot P is positioned in the vicinity of the boundary between the face side **11***a* and the face **21***a*.
[0199] Then, the chuck table **54** and the holding plate **42** are turned through the same angle per unit time, removing the region of the resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other by way of ablation circumferentially along the device wafer **11**.
[0200] Although not illustrated, in separation preparing step S**50**, the laser processing apparatus **52** may be used to perform an annular groove forming step in the same manner as with the fourth embodiment (see FIG. **17**). An example of laser processing conditions in the annular groove forming step is given below. The focused spot P may make one revolution or a plurality of revolutions around the rotational shaft **54***b* when the chuck table **54** and the holding plate **42** are turned through the same angle per unit time. [0201] Wavelength of laser beam: 355 nm to 532 nm [0202] Average output power: 6 W [0203] Repetitive frequency: 10 kHz [0204] Angle through which focused spot is turned per unit time around rotational shaft: 120°/s
Sixth Embodiment
[0205] A sixth embodiment of the present invention will be described below with reference to FIGS. **19**, **20**, and **21**. The sixth embodiment is the same as the first embodiment with regard to recess forming step S**10** through affixing step S**30**. In thinning step S**40**, however, a process called “TAIKO” (registered trademark) is carried out. The process will hereinafter be referred to as the “TAIKO process.”
[0206] In thinning step S**40** according to the sixth embodiment, not the entire reverse side **11***b* of the device wafer **1**, but only a central portion **17***f* (see FIG. **19**) of the reverse side **11***b* that is commensurate with the device region **17***a* of the face side **11***a* is thinned down (see FIG. **20**).
[0207] As illustrated in FIG. **19**, a grinding and polishing apparatus **22** has a grinding unit **64** for carrying out the TAIKO process, instead of the grinding unit **26***b* and the polishing unit **34**. The grinding unit **64** has a cylindrical spindle **66** whose longitudinal central axis extends vertically along the Z-axis.
[0208] A rotary actuator, not depicted, such as an electric motor is coupled to a longitudinally central portion of the spindle **66**. The spindle **66** has a lower end to which a disk-shaped wheel mount **68** is fixed. The wheel mount **68** has a lower surface on which an annular grinding wheel **70** is mounted.
[0209] The grinding wheel **70** has an annular base **70***a* of metal that has a lower surface on which there is mounted an annular array of grindstones **70***b* spaced at substantially equal angular intervals circumferentially along the base **70***a*. The grindstones **70***b* are arranged in an annular pattern having such a size that when the grinding wheel **70** is rotated about the central axis of the spindle **66**, the outside diameter of an annular path followed by the grindstones **70***b* is substantially half of the diameter of the device region **17***a*.
[0210] The holding surface **24***a* of the chuck table **24** according to the sixth embodiment is of an upwardly protruding conical shape. However, a chuck table, not depicted, including an outer circumferential portion and a central portion protruding upwardly beyond an annular region between the outer circumferential portion and the central portion and having a dual-recess cross-sectional shape may be used.
[0211] FIG. **19** illustrates thinning step S**40** according to the sixth embodiment, in side elevation, partly in cross section. In thinning step S**40** according to the sixth embodiment, the face **21***b* of the support plate **21** is held under suction on the holding surface **24***a* such that the reverse side **11***b* of the device wafer **11** is exposed upwardly.
[0212] Then, after the chuck table **24** has been placed directly below the grinding unit **64**, the rotational shaft **24***b* and the spindle **66** are rotated in respective predetermined directions, and the grinding unit **64** is lowered, i.e., grinding-fed, along the Z-axis at a predetermined speed.
[0213] During the grinding process, while an area where the grindstones **70***b* and the reverse side **11***b* are kept in contact with each other is supplied with grinding water such as pure water, the central portion **17***f* of the reverse side **11***b* is ground to thin down the device wafer **11**. In this

manner, a ring-shaped stiffener **17***g* is formed on an outer circumferential portion that is commensurate with the outer circumferential edge portion **17***b* (see FIG. **20**). An example of grinding conditions in thinning step S**40** is given below. [0214] Rotational speed of grinding wheel: 4000 rpm [0215] Rotational speed of chuck table: 300 rpm [0216] Grinding feed speed: 0.8 μm/s [0217] Flow rate of grinding water: 5.0 L/min

[0218] After the thickness of the central portion **17***f* of the device wafer **11** has reached a desired value, the grinding unit **64** is lifted. FIG. **20** illustrates, in side elevation, partly in cross section, the stack **25** that has gone thinning step S**40** according to the sixth embodiment. In thinning step S**40** according to the sixth embodiment, the coarse grinding process and the fine grinding process may be carried out. After the grinding process, the central portion **17***f* of the reverse side **11***b* may be polished.

[0219] According to the sixth embodiment, thinning step S**40** is followed by separation preparing step S**50** in which the resin removing step is carried out using the cutting apparatus **2** described above. FIG. **21** illustrates, in side elevation, partly in cross section, the resin removing step in separation preparing step S**50** according to the sixth embodiment.

[0220] In the resin removing step, the support plate **21** is held under suction on the chuck table **4**, and the holding plate **42** is inserted into a disk-shaped space positioned radially inwardly of the ring-shaped stiffener **17***g*, after which the holding plate **42** holds the central portion **17***f* of the reverse side **11***b* under suction thereon.

[0221] While the lower end **10***b* of the cutting edge **10***a* of the cutting blade **10** is positioned at a third depth C**3** extending from the reverse side **11***b* of the ring-shaped stiffener **17***g*, the region of the cured resin **19** that contributes to the affixing of the device wafer **11** and the support plate **21** to each other is removed circumferentially around the device wafer **11** by the cutting blade **10**.

[0222] At this time, as described above, the timings of starting to rotate the chuck table **4** and the holding plate **42** are synchronized, and the chuck table **4** and the holding plate **42** are rotated at the same rotational speed. In separation preparing step S**50** according to the sixth embodiment, the bonding force with which the device wafer **11** and the support plate **21** are bonded to each other along the thicknesswise direction **25***a* of the stack **25** is now reduced.

[0223] Instead of making the chopper cut, the chuck table **4** may be moved along the X-axis while the lower end **10***b* of the rotating cutting edge **10***a* is fixed in position along the Z-axis, causing the cutting edge **10***a* to cut to a desired position in the roll-off region **23***f*, after which the chuck table **4** may start rotating about its vertical central axis.

[0224] According to the sixth embodiment, as with the first modification (see FIG. **12**) of the first embodiment, the chuck table **4** may hold the device wafer **11** under suction thereon, and the holding plate **42** may hold the support plate **21** under suction thereon. In this case, the diameter of the chuck table **4** is made smaller than the inside diameter of the ring-shaped stiffener **17***g*, and the chuck table **4** inserted into the disk-shaped space positioned radially inwardly of the ring-shaped stiffener **17***g* holds the central portion **17***f* of the reverse side **11***b* under suction thereon.

[0225] Moreover, as with the second modification (see FIG. **13**) of the first embodiment, the recess **21***d* may be expanded radially outwardly of the support plate **21** to form an annular step whose outer circumferential portion is open radially outwardly in the outer circumferential region **23***b* of the support plate **21** on the face **21***a* thereof.

[0226] Furthermore, as with the second embodiment (see FIG. **14**A), the recess **17***d* may be formed in the face side **11***a* of the device wafer **11**, and as with the third embodiment (see FIG. **16**A), the recess **17***d* may be formed in the face side **11***a* of the device wafer **11** and the recess **21***d* may be formed on the face **21***a* of the support plate **21**.

Modification of Support Plate

[0227] A modification of the support plate **21** will be described below with reference to FIGS. **22**A and **22**B. FIG. **22**A illustrates, in plan, the support plate **21** according to the modification, and FIG. **22**B is a cross-sectional view taken along line B-B of FIG. **22**A.

[0228] The support plate **21** according to the modification includes a substrate of monocrystalline silicon and includes a plurality of substantially cylindrical pores **21**_e_ extending therethrough from the central region **23**_a_ of the face **21**_a_ along the thicknesswise direction **21**_c_ to the face **21**_b_. Each of the pores **21**_e_ may be formed by way of laser ablation, for example.
[0229] The pores **21**_e_ have a diameter in a range from 1 μm to 20 μm, which is set to a predetermined value in a range from 10 μm to 20 μm, for example. However, the diameter of the pores **21**_e_ is not limited to the value mentioned. Each of the pores **21**_e_ may not necessarily be of a circular shape on the face **21**_a_. Each of the pores **21**_e_ may be shaped as a right prism having a polygonal bottom surface or may be shaped otherwise.
[0230] The pores **21**_e_ are arranged in a regular pattern on the face **21**_a_. As illustrated at an enlarged scale in a right inset in FIG. **22**A, two closest pores **21**_e_ in a first direction **21**_f_ on the face **21**_a_ are spaced from each other by (predetermined distance D)×2, and two closest pores **21**_e_ in a second direction **21**_g_, which is perpendicular to the first direction **21**_f_, on the face **21**_a_ are also from each other by (predetermined distance D)×2. For example, the predetermined distance D represents 5.0 mm.
[0231] Stated otherwise, square shapes **21**_h_ whose each side has a size of (predetermined distance D)×(√2) are closely packed in the central region **23**_a_ of the face **21**_a_, with pores **21**_e_ positioned at the respective vertexes of each square shape. In other words, the positions of the pores **21**_e_ thus correspond to the grid points of unit grids that are two-dimensionally packed in the central region **23**_a_. The pores **21**_e_ are not limited to the pitch and layout illustrated in FIG. **22**A.
[0232] As illustrated in FIG. **22**B, a modified region **21**_i_ where the crystal property of monocrystalline silicon has been changed is formed around each of the pores **21**_e_ so as to extend from the face **21**_a_ to the face **21**_b_ in surrounding relation to the outside of the pore **21**_e_.
[0233] The modified region **21**_i_ refers to a region where the crystal structure and density of part of the support plate **21** have been changed by receiving the energy from a laser beam. In FIG. **22**B, the modified regions **21**_i_ are depicted as stippled, distinguishing themselves from regions depicted as hatched, i.e., regions where the crystal structure and density have not been changed.
[0234] Each of the modified regions **21**_i_ includes an amorphous region or a polycrystalline region. Each of the modified regions **21**_i_ has an inside diameter commensurate with the outside diameter of the pore **21**_e_ surrounded thereby. The outside diameter of each of the modified regions **21**_i_ represents 20 μm or less, for example.
[0235] For fabricating the support plate **21** according to the modification, a pulsed layer beam is applied to the support plate **21** to perform ablation thereon, for example, forming processed regions including pores **21**_e_ and modified regions **21**_i_ in the support plate **21**. An example of processing conditions is given below. [0236] Wavelength of laser beam: 1064 nm [0237] Pulse energy: 50 μJ [0238] Repetitive frequency of pulses: 1 kHz [0239] Processing feed speed: 20 mm/s
[0240] The diameter, pitch, and layout of the pores **21**_e_ may appropriately be varied by changing the processing conditions. In addition to ablation, etching may further be applied to the support plate **21** with an alkaline etching solution, partially removing the modified regions **21**_i_ to increase the diameter of the pores **21**_e_. The diameter of the pores **21**_e_ can be increased by adjusting the amount of the removed portion of the modified regions **21**_i_.
[0241] In affixing step S**30**, when the face **21**_a_ of the support plate **21** that has the pores **21**_e_ and the face side **11**_a_ of the device wafer **11** are affixed to each other, gas that is positioned between the face **21**_a_ and the face side **11**_a_ can be discharged through the pores **21**_e_.
[0242] Therefore, when the device wafer **11** is thinned down in thinning step S**40**, variations of the thickness of the device wafer **11** can be reduced and the thickness of the device wafer **11** can be uniformized with higher accuracy than if gas remained as gas bubbles between the face **21**_a_ and the face side **11**_a_.
[0243] The support plate **21** with the pores **21**_e_ defined therein is not limited to the substrate of monocrystalline silicon but may be made of a material such as ceramic, glass, resin, or metal, for

example, other than monocrystalline silicon.

[0244] The structural and methodical details of the embodiments described above may be changed or modified without departing from the scope of the invention. According to the above embodiments, the recess **17***d* in the device wafer **11** and the recess **21***d* in the support plate **21** includes single continuous annular slots or steps.

[0245] However, the recess **17***d* may be a plurality of discrete, i.e., unconnected and independent, slots disposed circumferentially along the face side **11***a*. Similarly, the recess **21***d* may also be a plurality of discrete slots or steps disposed circumferentially along the face **21***a*.

[0246] In a case where the resin **19** includes a thermosetting resin, the stack **25** may be heated to reduce the bonding force with which the device wafer **11** and the support plate **21** are bonded to each other by the resin **19** in separation preparing step S**50**, rather than being cut by the cutting blade **10** and ablated by the laser beam L.

[0247] The present invention is not limited to the details of the above described preferred embodiments. The scope of the invention is defined by the appended claims and all changes and modifications as fall within the equivalence of the scope of the claims are therefore to be embraced by the invention.

## Claims

**1**. A processing method of processing a device wafer having on a face side thereof a device region where devices have been formed and an excess outer circumferential region surrounding the device region, the method comprising: forming a recess circumferentially along a support plate including a face having a size equal to or larger than a diameter of the device region and supporting the device wafer, in an outer circumferential region of the face that is positioned outwardly of a central region thereof that is commensurate with the device region, or circumferentially along the device wafer in the excess outer circumferential region of the face side of the device wafer; supplying the recess with resin; affixing the face of the support plate and the face side of the device wafer to each other with use of at least the resin, thereby forming a stack; thinning down the device wafer by grinding or polishing or by grinding and polishing a reverse side of the device wafer that is positioned opposite the face side of the device wafer in a thicknesswise direction thereof; removing a region of the resin that contributes to the affixing of the device wafer and the support plate to each other circumferentially along the device wafer, thereby to reduce a bonding force with which the device wafer and the support plate are bonded to each other in a thicknesswise direction of the stack, or forming an annular slot in the device wafer that extends from the face side to the reverse side thereof outwardly of the device region and inwardly of an annular region aligned with the recess, thereby to sever a portion of the device wafer that includes the device region from the support plate; and separating the portion of the device wafer that includes the device region from the support plate.

**2**. The processing method according to claim 1, further comprising: before the stack is formed, forming an additional recess circumferentially in the excess outer circumferential region or the outer circumferential region, which is free of the recess, of the face of the support plate and the face side of the device wafer; and before the stack is formed, supplying the additional recess with an additional resin.

**3**. The processing method according to claim 1, wherein, in supplying the recess with the resin, while a lower end of a cutting blade is positioned at a first depth from the reverse side of the device wafer or at a second depth from another face of the support plate that is positioned opposite the face of the support plate in a thicknesswise direction thereof, the region of the resin that contributes to the affixing of the device wafer and the support plate to each other is removed by the cutting blade circumferentially along the device wafer, thereby reducing the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.

**4**. The processing method according to claim 1, wherein, in supplying the recess with the resin, a pulsed layer beam having a wavelength absorbable by the device wafer is applied from the reverse side of the device wafer to another face of the support plate that is positioned opposite the face of the support plate in a thicknesswise direction thereof or from the other face of the support plate to the reverse side of the device wafer, thereby to remove the region of the resin that contributes to the affixing of the device wafer and the support plate to each other circumferentially along the device wafer, thereby reducing the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.

**5**. The processing method according to claim 1, wherein, in thinning down the device wafer, a central portion of the reverse side of the device wafer that is commensurate with the device region is thinned down to form a ring-shaped stiffener on an outer circumferential portion thereof that is commensurate with the excess outer circumferential region, and, in reducing the bonding force with which the device wafer and the support plate are bonded to each other or in severing the portion of the device wafer from the support plate, the region of the resin that contributes to the affixing of the device wafer and the support plate to each other is removed circumferentially along the device wafer to reduce the bonding force with which the device wafer and the support plate are bonded to each other in the thicknesswise direction of the stack.

**6**. The processing method according to claim 1, wherein, while the support plate is held on a holding table and the device wafer is held on a holding plate different from the holding table, the bonding force with which the device wafer and the support plate are bonded to each other is reduced or the portion of the device wafer is severed from the support plate.

**7**. The processing method according to claim 1, wherein the support plate includes a plurality of pores extending therethrough from the central region of the face thereof along a thicknesswise direction of the support plate to another face of the support plate that is positioned opposite the face thereof, and, in forming the stack, the face of the support plate and the face side of the device wafer are affixed to each other.